

Using Wavefront Tracing for the Visualization and Optimization of Progressive Lenses

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Abstract

Progressive addition lenses are a relatively new approach to compensate for defects of the human visual system. While traditional spectacles use rotationally symmetric lenses, progressive lenses require the specification of free-form surfaces. This poses difficult problems for the optimal design and its visual evaluation.

This paper presents two new techniques for the visualization of optical systems and the optimization of progressive lenses. Both are based on the same wavefront tracing approach to accurately evaluate the refraction properties of complex optical systems.

We use the results of wavefront tracing for continuously re-focusing the eye during rendering. Together with distribution ray tracing, this yields high-quality images that accurately simulate the visual quality of an optical system. The design of progressive lenses is difficult due to the trade-off between the desired properties of the lens and unavoidable optical errors, such as astigmatism and distortions. We use wavefront tracing to derive an accurate error functional describing the desired properties and the optical error across a lens. Minimizing this error yields optimal free-form lens surfaces.

While the basic approach is much more general, in this paper, we describe its application to the particular problem of designing and evaluating progressive lenses and demonstrate the benefits of the new approach with several example images.

1. Introduction

In order to sharply see objects at different distances, the lens of the human eye has to alter its curvature and thus its power of refraction, i.e. the eye *accommodates*. Due to gradual stiffening of the lens with age, the ability to accommodate decreases. This kind of visual defect, known as *presbyopia*, can be corrected by *progressive-addition lenses (PAL)*.

A PAL has two areas, one with low refractive power for viewing objects at far distances and one with higher refractive power for objects closer to the eye. Thus, the required accommodation of the eye is (partially) supplied by an increasing refractive power of the correcting lens. Usually, a large *far vision area* is specified for the upper part of a PAL, and a smaller *near vision area* for the central lower parts.

Furthermore, it is important for the visual quality of a lens to have a smooth transition of refractive powers across the PAL (see Figure 1).

Color Figure 17 shows on the right hand side the color-coded refractive power and optical error (astigmatism) of two different well-designed PALs. Unfortunately, some error is unavoidable for this kind of lenses, but for an optimal design the error is pushed towards the lower left and right periphery of the lens and is smoothly distributed. In particular, errors in the far and near vision area and along the main viewing line should be minimal for a good design.

In this paper, we describe both a new technique for visualizing the visual quality of optical systems and an optimization technique for optical lenses. Both depend on *wavefront tracing* in order to evaluate the exact properties of the optical system. In contrast to previously used coarse approximations, wavefront tracing allows for much more accurate simulations and consequently better lenses.

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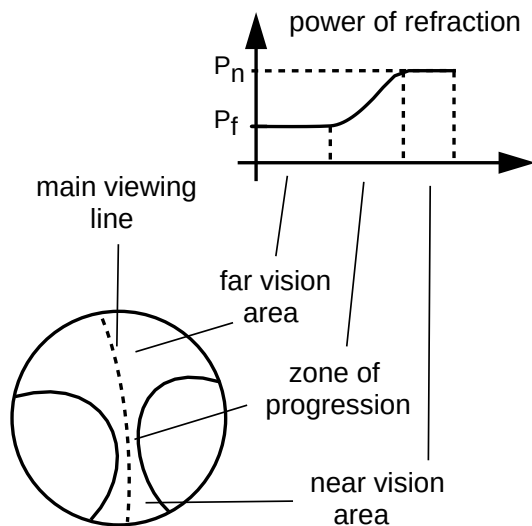


Figure 1: Typical prescribed distribution of refractive power on a PAL

For demonstrating the visual properties of optical systems we render images of 3D scenes as they would be seen by a human observer when looking through the lens and trying to focus on each visible object individually. A specially designed rendering algorithm uses wavefront tracing^{1, 2} for re-focusing the eye for each pixel and uses distribution ray tracing³ for sampling the visible objects. Thus, the algorithm allows us to simulate depth of field effects, image distortions and, in contrast to previous approaches^{4, 5, 6, 7, 8}, the effects of accommodation of the human eye.

We also describe a new method for computing the geometry of PALs that minimizes optical errors as much as possible and gives the PAL-designer the flexibility to create PALs with prescribed optical properties. Our method is based on combining wavefront tracing and variational design^{9, 10}.

We start our presentation in Section 2 with a brief review of the relevant optical factors of spectacle lenses and visual defects. This can best be understood through the concept of *wavefronts* and we describe how power and astigmatism of a lens is computed by performing wavefront tracing of the lens.

In Section 3, we describe how the view of a human eye can be simulated with ray tracing. For each pixel, wavefront tracing computes the required accommodative effort of the human eye in combination with an optical system. Distribution ray tracing is used for computing the final image. In order to compute the required accommodation of the eye, we present a model of the human eye, that is accurate enough to take into account the relevant physiological parameters, yet simple enough to be evaluated quickly, resulting in fast rendering and optimization algorithms.

In Section 4, we develop an error functional that is based on wavefront tracing and the new notion of *effective astigmatism*, a generalization of traditional error measures used in optics.

Finally in Section 5, we describe an efficient method for minimizing this error functional, thereby optimizing the PAL. Our method is based on a carefully chosen representation of the PAL surfaces. This representation enables efficient wavefront tracing and thus a quick numerical evaluation of the error functional. The minimization is done by approximating the functional with data dependent quadratic functionals.

Our method has been developed and tested in close cooperation with industry and is currently introduced into the design process. The results shown in this paper demonstrate its superiority over previous approaches.

1.1. Previous work

A fast but severely restricted approach to obtain optical effects like depth of field are post-filtering methods^{4, 5}. In contrast, distribution ray tracing³ has been used extensively for accurately rendering effects such as depth of field and optical distortions. This approach has also been used for the realistic simulation of complex optical systems⁶. However, neither technique offers means to adjust and focus a non-trivial optical system (like combination of the human eye and a lens).

Wavefront tracing is best known in computer graphics as a tool for illumination computations^{11, 12, 2, 13}. In this work, we use wavefront tracing in a different context by evaluating the wavefronts for focusing a lens system and computing its optical properties.

One of the first PALs was manufactured by Maitenaz¹⁴ in 1967. His approach is based on simple heuristics and offers only a few free parameters for optimizing the design. The surface of the lens is obtained by moving a conic section along a curve with increasing curvature. Since then more general free-form surfaces have been used for describing the geometry of PALs^{15, 16, 17}. Unfortunately these approaches do not take into account the exact optical quantities of the lens, but rather use a *thin lens approximation*, thus trading accuracy for faster computations. With our new wavefront-based approach we are able to exploit the exact optical quantities, while still using fast simulation and optimization algorithms.

2. Visual Defects, Wavefronts and Spectacle Lenses

Before describing the new approach, we give a brief overview of visual defects that can be corrected by spectacle lenses and describe the basic characteristics of lenses by means of wavefronts.

2.1. Visual Defects and Their Correction by Spectacle Lenses

An eye is said to be correctly sighted if a point at infinity gets mapped sharply on the retina using the least accommodation effort by the eye, i.e. when the lens is at rest. In optical terms a thin bundle of parallel light rays along the optical axis is deflected and should converge to the same point on the retina (see Figure 2a). This area, which only considers points at infinity and assumes an unflexed lens is called *Ophthalmology*.

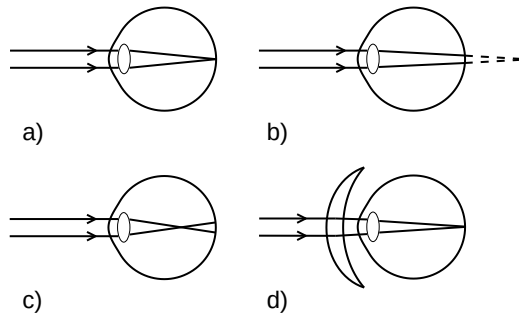


Figure 2: Visual defects and their correction: a) correctly sighted eye, b) far sighted eye, c) short sighted eye, d) far sighted eye with correcting lens.

An eye is *short sighted* (*far sighted*), if it has its focal point in front of (behind) the retina (Figure 2b + c). For correcting these defects, a concave (convex) ophthalmic lens is used (Figure 2d). Most commonly, the front and back surfaces of the lens have spherical geometry.

If the bundle of rays does not intersect in a single point after refraction, the eye is called *astigmatic*. This visual defect is caused by a non-symmetrical eye structure. A first order approximation assumes a toroidal lens surface with two different principal curvatures, which result in two perpendicular *focal lines* instead of a single focal point (see Figure 3). Depending on the position of the retina relative to the focal lines, far sighted astigmatism, mixed astigmatism and short sighted astigmatism are distinguished. Due to astigmatism, the eye is unable to sharply view objects at any distance. A correcting spectacle lens usually has a spherical front surface and a toroidal back surface (Figure 4), where the torus is oriented with respect to the orientation of the focal lines.

An eye is called *presbyopic*, if its accommodation ability is significantly restricted. For correcting this defect, a PAL, i.e. a lens with varying power of refraction is required. Since the power of refraction mostly depends on the mean curvature of the front surface, it is replaced by a free form surface that exhibits increasing mean curvature from the far vision to the near vision area.

An additional non-astigmatic or astigmatic visual defect can be corrected by a spherical or toroidal back face as

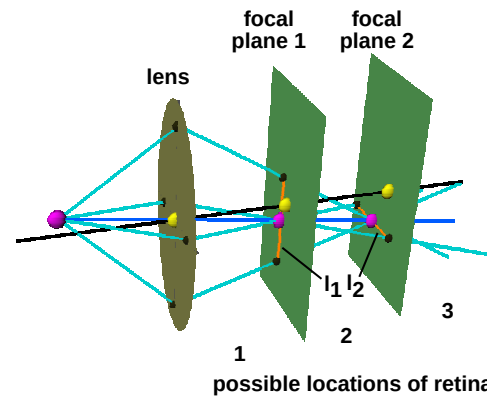


Figure 3: Different forms of astigmatism depending on the position of the retina relative to the focal lines (l_1 and l_2): far sighted (1), mixed (2), near sighted astigmatism (3).

described above. The restriction to these simple back surfaces is due to cost and manufacturing constraints of creating PALs. Usually, a selection of pre-manufactured moulded front surfaces is made available, which can then be adapted locally by an optician to the requirements of a particular patient by cutting a simple spherical or toroidal back face.

2.2. Characterizing Ophthalmic Lenses by Wavefronts

The main characteristics of ophthalmic lenses are *power of refraction* and *astigmatism*. We first consider the simpler case of rotationally symmetric glasses as in Figure 5.

After refraction a parallel bundle of rays intersects the optical axis at the *focus* F . The *focal length* is $\overline{BF}$, the directed distance from F to B with respect to the optical

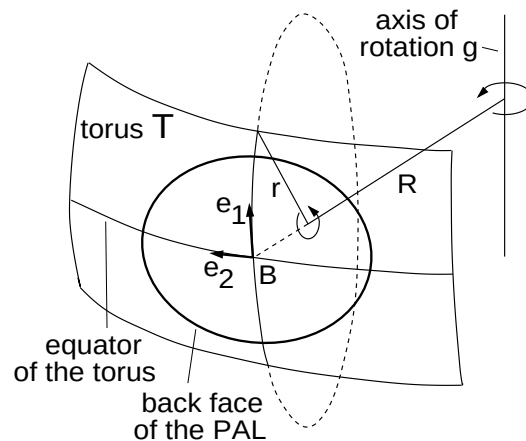


Figure 4: Using a toroidal surface as the back face of a PAL. e_1, e_2 are the directions of principal curvatures.

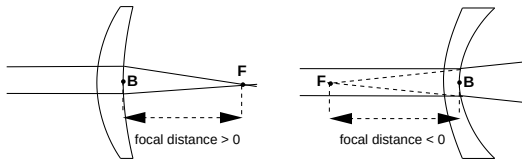


Figure 5: Power of refraction for rotationally symmetric lenses B: reference point at the back of the lens, F: focus point.

flow. Thus, $\overline{BF}$ is negative for concave lenses. The reciprocal value $P = 1/\overline{BF}$ is called the *power of refraction*. It is specified in *dioptr* = dpt [m^{-1}].

The above characterization obviously fails for lenses with arbitrary geometry but also for off-axis light rays. In order to extend the term *refractive power* to arbitrary cases and to introduce *astigmatism*, the concept of *wavefronts*^{1, 18, 19} is required. Consider a point light source that is switched on momentarily at the time t_0 . At the time $t > t_0$ the photons emitted at t_0 form a surface W_t , the wavefront. When passing through an inhomogeneous medium or due to refraction, the shape of the wavefront W_t changes.

Formally, we parameterize the wavefront as a time-dependent surface $W(u, v, t) = W_t(u, v)$. In the following, we are interested in the local curvatures of W_t along a particular light ray $r(u_0, v_0) : t \mapsto W(u_0, v_0, t)$. The generalization of refractive power using local wavefront curvatures is quite evident: We choose the *back vertex* B on the lens' back face, indicating the main viewing direction and serving as a point of reference. The center of rotation of the viewer's eye will be denoted by C and we construct a *vertex sphere* V centered at C that intersects B (see Figure 6).

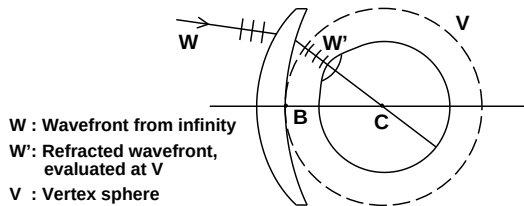


Figure 6: Optical model for the general definition of power and astigmatism through the curvature of wavefronts.

We now consider a point at infinity viewed by the eye. This point emits a ray $W(u_0, v_0, t)$, corresponding to a planar wavefront $W(u, v, t)$. After refraction, the ray intersects V at the time $t = t_1$. The principal (minimum and maximum) curvatures κ'_1, κ'_2 of the corresponding refracted wavefront W'_{t_1} evaluated at (u_0, v_0) , serve for defining the (*vertex*) *power* P and the (*vertex*) *astigmatism* A along the considered viewing

direction:

$$\begin{aligned} P &= \frac{\kappa'_1 + \kappa'_2}{2} \\ A &= \kappa'_1 - \kappa'_2 \end{aligned} \quad (1)$$

2.3. Transformations of Wavefront Curvatures

In order to compute power and astigmatism with wavefronts, we need to know how wavefronts propagate. The transformation formulas describe how the wavefront's curvatures change when passing through an optical system. Two cases are relevant for ophthalmic lenses:

- Spread in a homogeneous medium,
- Transition at an optical boundary.

The respective transformation formulas and their derivation are well-known^{1, 2}. Spread in homogeneous media is described by $\kappa_i(t) = \frac{\kappa_i}{1 - ct\kappa_i}$, ($i = 1, 2$), where κ_1, κ_2 are the wavefront curvatures at $t = 0$ and c is the speed of light (see Figure 7).

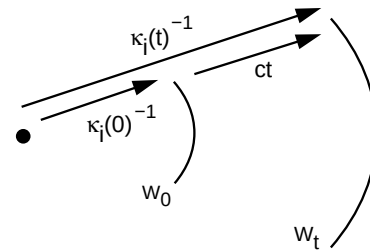


Figure 7: Curvature transformation for spread of a wavefront in a homogeneous medium

In the case of refraction, we have to transform the wavefront's normal N , the principal curvatures κ_1, κ_2 and the corresponding principal directions e_1, e_2 . The formulas^{1, 2} allow us to directly compute the quantities N', κ'_1, κ'_2 and e'_1, e'_2 of the refracted wavefront W' as a function of the corresponding curvature of the incident wavefront W and the refracting surface S . See the references for a more detailed derivation of the transformation.

For refraction, consider a wavefront $W(u, v, t)$, passing along a ray $W(u_0, v_0, t)$ and being refracted by the surface S . We adopt the convention that if we do not specify (u, v, t) explicitly, then we are referring to $(u, v, t) = (u_0, v_0, 0)$, where $t = 0$ is the time when $W(u_0, v_0, t)$ hits the surface.

First we have to compute the normal N' to the refracted wavefront W' . We need a representation of N' as a linear combination $N' = \mu N + \gamma \hat{N}$, with N being the normal vector of W before refraction and $\hat{N}$ being the surface normal. Using Snell's law, it is easy to derive $\mu = \frac{n_1}{n_2}, \gamma = -\mu \cos \varphi + \cos \varphi'$. Here, n_1 and n_2 are the indices of refraction of the media before and behind the surface S , respectively. φ and

φ' denote the angle of incidence and the angle of refraction, respectively.

Then we choose a common unit tangent vector ξ of W and S , and a unit vector η on the tangent plane of S , which is perpendicular to ξ . Next, we determine the angle θ between ξ and the first principal direction e_1 on W , i.e.,

$$\xi = \cos \theta e_1 - \sin \theta e_2, \quad \eta = \sin \theta e_1 + \cos \theta e_2 \quad (2)$$

and apply Euler's formulas^{1,2}

$$\begin{aligned} \kappa_\xi &= \kappa_1 \cos^2 \theta + \kappa_2 \sin^2 \theta \\ \kappa_\eta &= \kappa_1 \sin^2 \theta + \kappa_2 \cos^2 \theta \\ \kappa_{\xi\eta} &= (\kappa_1 - \kappa_2) \cos \theta \sin \theta \end{aligned} \quad (3)$$

to find the directional curvatures κ_ξ , κ_η and $\kappa_{\xi\eta}$ on W , where $\kappa_{\xi\eta}$ is the torsion of the geodesic in direction ξ . We compute the directional curvatures $\hat{k}_\xi$, $\hat{k}_\eta$ and $\hat{k}_{\xi\eta}$ on S analogously.

The following transformation formulas^{1,2} allow us to compute the corresponding curvatures κ'_ξ , κ'_η and $\kappa'_{\xi\eta}$ on W' :

$$\begin{aligned} \kappa'_\xi &= \mu \kappa_\xi + \gamma \hat{k}_\xi \\ \kappa'_\eta &= \mu \kappa_\eta \frac{\cos^2 \varphi}{\cos^2 \varphi'} + \gamma \hat{k}_\eta \frac{1}{\cos^2 \varphi'} \\ \kappa'_{\xi\eta} &= \mu \kappa_{\xi\eta} \frac{\cos \varphi}{\cos \varphi'} + \gamma \hat{k}_{\xi\eta} \frac{1}{\cos \varphi'} \end{aligned} \quad (4)$$

Inversion of Euler's formula yields the principal curvatures κ'_1 , κ'_2 and the angle θ' which allows us to compute the principal directions e'_1 and e'_2 on W' . Thus we determine the curvature information of W' using the corresponding terms of W and S .

3. Simulating the Vision of the Human Eye

While the optical properties of a lens can be visualized by color coding the power of refraction and astigmatism (see Figure 17), its effect on the human vision cannot be derived from these color plots. Instead, we developed an algorithm that visualizes the combined optical system composed of eye and lens by rendering images of 3D-scenes.

Previous rendering techniques used a fixed optical system for rendering images⁶. However, this means that for many scenes most parts of the image would be blurred because the visible objects are out of focus. Instead, we want to visualize the effect of the lens over the full field of view. Thus, we used the fact that for each direction a human observer would focus on the object in the line of sight. Similarly, we re-focus the virtual eye for every pixel before computing its intensity. As a result, we obtain an image that would be perceived by a viewer as he scans the field of view. Re-focusing is performed based on our model of the human eye and the data obtained by wavefront tracing.

3.1. Eye Model

We assume that the eye has a spherical shape and that the refractive power is completely provided by a thin lens, which is located at the pupil (see Figure 8). For our purposes, this approximation is accurate enough to model the visual defects, including far and short sightedness, astigmatism and presbyopia with the required accuracy, since we only consider light rays that are close to the optical axis. On the other hand, the model is still simple enough for efficient evaluation.

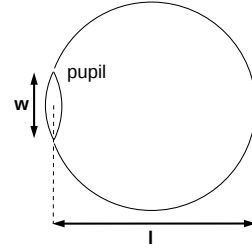


Figure 8: Simplified eye model.

The geometry of the eye can be specified by the diameters l and pupil widths w , for which the Gullstrand eye¹⁶, a standard eye model frequently used in ophthalmology, provides standard values ($l = 24$ mm, $w = 3.6$ mm). For rendering a rectangular image, we sample the field of view at regularly spaced pixels on a plane at the back of the eye and orthogonal to the main viewing direction.

Consider a lens with principal curvatures $\hat{k}_1^f, \hat{k}_2^f$ and $\hat{k}_1^b, \hat{k}_2^b$ at its front face and back face, respectively, and a relative index of refraction $n = n_1/n_2$. In ophthalmic optics

$$D^f := (n-1) \frac{\hat{k}_1^f + \hat{k}_2^f}{2}, \quad D^b := (1-n) \frac{\hat{k}_1^b + \hat{k}_2^b}{2}$$

are called *surface powers of refraction*, whereas the values

$$C^f := (n-1)(\hat{k}_1^f - \hat{k}_2^f), \quad C^b := (n-1)(\hat{k}_1^b - \hat{k}_2^b)$$

are known as *surface astigmatisms*.

Using the above wavefront-transformations, it is easily derived that power and astigmatism of a thin lens are given by

$$P = D^f + D^b, \quad A = C^f - C^b. \quad (5)$$

An important special case, which we require later for developing an appropriate eye model, is the orthogonal refraction of a wavefront by a thin lens. Although the transformations for the wavefront curvatures κ'_1, κ'_2 remain complicated in this situation, the transformation of the mean curvature is quite simple:

$$\frac{\kappa'_1 + \kappa'_2}{2} = \frac{\kappa_1 + \kappa_2}{2} + P \quad (6)$$

3.1.1. Computing Accommodation

In order to obtain a functional eye model, we have to specify how the eye accommodates when looking at a wavefront W' . We also need to specify the astigmatism of the eye and the range of possible accommodation.

We denote the two principle refractive powers of the eye with least accommodation by $Q_{\min,1}$ and $Q_{\min,2}$. When looking at objects from short distances, the refractive powers increase $Q_i = Q_{\min,i} + \Delta Q$, but the accommodative effort ΔQ is limited to the range $\Delta Q \in [0, \Delta_{\max}]$. The maximal accommodation ability $\Delta_{\max}$ depends heavily on age as is shown in Figure 9.

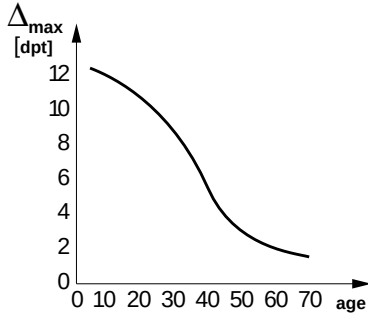


Figure 9: Average loss of accommodation with age²⁰.

Next, we compute the required accommodation effort ΔQ when looking at a wavefront W' . We denote the wavefront after refraction at the eye by W'' . For the general case of an astigmatic eye, one focal line of W'' will be located before, the other behind the retina after accommodation. We adopt the common assumption, that the eye tries to accommodate so as to create a smallest circle of confusion on the retina. This means that $r_1 = r_2$ in Figure 10. Then it can be derived by simple geometric considerations, that the curvatures have to fulfill $(\kappa'_1 + \kappa'_2)/2 = 1/l$.

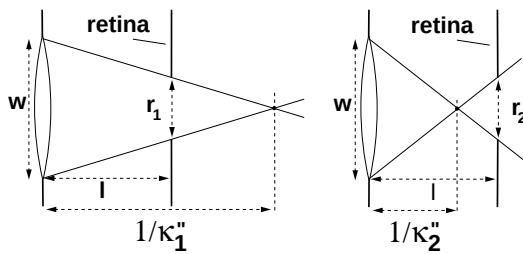


Figure 10: Astigmatic wavefronts have two focal lines generating an ellipse of confusion on the retina. However, sufficient accommodation ability provided, a circle of confusion can be obtained by properly positioning the focal lines before and after the retina so that $r_1 = r_2$.

From Equation 6, we conclude that

$$Q = \frac{1}{l} - \frac{\kappa'_1 + \kappa'_2}{2} \quad (7)$$

is the required refractive power of the eye, and thus

$$\Delta Q = \frac{1}{l} - \frac{\kappa'_1 + \kappa'_2}{2} - \frac{Q_{\min,1} + Q_{\min,2}}{2}. \quad (8)$$

Taking into account the restricted accommodation ability, we finally set $\Delta Q := \max(\min(\Delta Q, \Delta_{\max}), 0)$.

3.2. The Ray tracing Algorithm

Now that we know how to compute the accommodation of the human eye, we render the simulated view as outlined above. The rendering algorithm performs three tasks for each pixel (see Figure 11).

3.2.1. Locating the Visible Object

First of all, we need to determine the point in the environment on which the eye should focus. This is done by tracing a single ray from the center of the pixel through the center of the pupil and through the lens into the scene determining the first intersection with any object (see Figure 11a).

3.2.2. Wavefront Tracing and Accommodation

The next step involves wavefront tracing and serves for computing the required accommodation effort of the eye. A spherical wavefront W is initiated at the point determined in step one. We trace the wavefront along the reverse direction of the ray. Using the curvature transformation formulas, we compute the curvatures of the incident wavefront W' at the pupil, before entering the eye (Figure 11b). In order to get a sharp image of the object point, the eye now accommodates according to the eye model described above.

3.2.3. Distribution Ray tracing

Now that the refractive power of the eye is determined, we keep it fixed and use distribution ray tracing³ for computing the final pixel color. For this purpose, we choose a set of uniformly distributed sample points on the pupil and the pixel, and trace eye rays into the environment. The final pixel color is obtained as a weighted average of the incident illumination (Figure 11c).

In the case of a non-astigmatic eye, a focal plane exists (see Figure 12). Its distance d from the aperture (the pupil), is the reciprocal value of the power of refraction. We use the standard procedure to obtain the direction of the ray: the object point X , corresponding to the image point (pixel) X' , is constructed by intersecting the principal ray with the focal plane, since the principal ray passes through the lens without being refracted. Every other sample ray is refracted so as to intersect the focal plane at the same object point X .

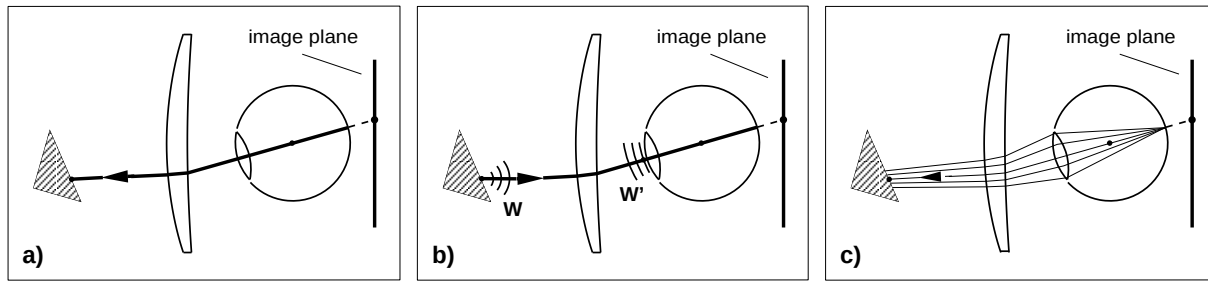


Figure 11: Visualizing the view through a spectacle lens: a) locating the object point with pin hole camera, b) determining the accommodative effort using wavefront tracing, c) computing the pixel color using distribution ray tracing.

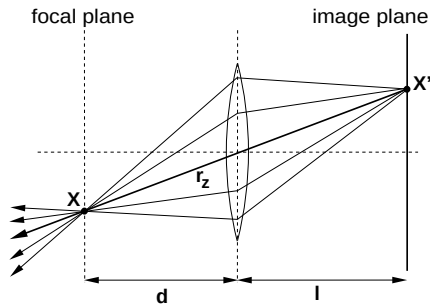


Figure 12: Tracing rays through a non-astigmatic thin lens

The construction for astigmatic lenses is a bit more complicated (see Figure 13). Essentially, we have to perform the same computation on two orthogonal planes for the two focal lines X_i . Their distance from the thin lens is given as the inverse of the refraction powers of the lens $1/P_i$, ($i = 1, 2$). Every sample ray is now characterized by intersecting both focal lines after refraction. The orientation of the focal lines corresponds to the astigmatism of the eye.

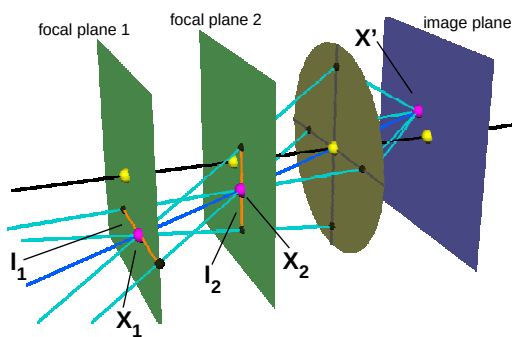


Figure 13: Tracing rays through an astigmatic thin lens

3.3. Examples

For computing the visual quality of a lens, we render images of several 3D environments. Special environments can be used to determine the optical characteristics of the lens. For instance, astigmatism can be visualized by rendering a scene having dotted textures, as the dots will be blurred to ellipses. The images in Figure 16 give a general impression of the optical effects and the image distortion due to different lenses.

The rendering algorithm has been implemented in the physically-based rendering architecture²¹ Vision. Thus, the approach can be applied to any kind of optical system that can be described in the RenderMan RIB format. We used CSG to model the spherical lenses and NURBS surfaces for describing PALs. The overhead for wavefront tracing is negligible compared to the cost of distribution ray tracing.

The image in the upper left of Figure 16 shows the view of a presbyopic eye, being short sighted with -1 dpt and a remaining accommodative ability of 0.5 dpt. In the upper right, a simple spectacle for near vision with a power of $+2$ dpt is used for correction for short distances (“reading glasses”). The image in the lower left shows the view through a -1 dpt lens, correcting vision for far distances (“driving glasses”). In either case, large parts of the image cannot be focused adequately onto the retina and are blurred excessively.

For the image in the lower right of Figure 16 a PAL lens is used to view the scene. The PAL has been designed by the optimized method described in the following sections. Good perception is achieved for all distances. However, near the lower left and right edges some optical aberrations become visible due to the unavoidable remaining astigmatism of the lens (compare to Figure 17). In addition, a very slight bending of the horizontal edge of the desk can be observed.

4. Characterizing a Progressive Lens by an Error Functional

Uni-focal lenses should have a constant power of refraction P and constant astigmatism A for the whole glass. In contrast, for PALs two different refractive powers P_F , P_N for

far and near vision area are required, with a constant astigmatism A for the whole glass. In this section, we develop an error functional, which describes the optimal progressive lens.

For a non-astigmatic eye, the lens should have vanishing astigmatism, i.e. $A = 0$. Therefore the error functional for non-astigmatic lenses is quite evident:

$$J(F) := \int_{\Xi} \alpha A^2 + \beta (P - P_{des})^2 d\phi d\theta. \quad (9)$$

P_{des} describes the desired distribution of refractive power over the lens. Using weight functions α and β , the glass designer can control how an occurrence of astigmatism or error in refractive power is penalized. The integrand is evaluated on the vertex sphere V and the integral is computed over all viewing directions, which are parameterized by angles ϕ , θ in polar coordinates. Usually, we prefer small astigmatism A in central areas of the lens and allow for larger errors near the edges.

The situation for astigmatic lenses is more complicated. We may keep the error term $(P - P_{des})^2$ in Equation 9 but we have to adapt the term A^2 . We cannot simply replace A^2 by something like $(A - A_{des})^2$, since this would ignore the orientation of the astigmatism. Therefore additional considerations are required.

After refraction, the wavefront W' is transformed to a wavefront W'' by the eye lens (Figure 14). For optimal correction the curvature term $A = \kappa'_1 - \kappa'_2$ vanishes, since only then a single focus point may occur. At a first glance, one could use $(\kappa'_1 - \kappa'_2)^2$ as a substitute for A^2 in Equation 9. However, this would not lead to a true generalization of Equation 9, since we need to evaluate the functional at the vertex sphere V .

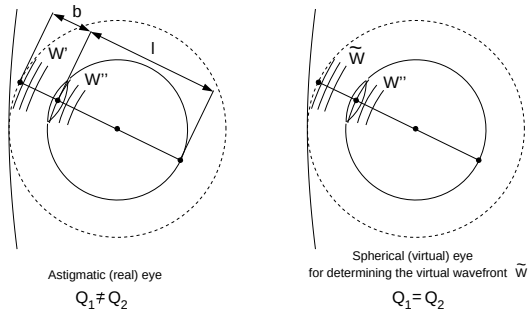


Figure 14: Characterizing the virtual wavefront. Refracting the virtual wavefront $\tilde{W}$ at a spherical, virtual eye (right) is equivalent to refracting the real wavefront W' at the real eye (left).

We therefore introduce the term *effective astigmatism* A_{eff} , which is a new concept for describing the optical performance of an astigmatic lens. Next, the astigmatic eye with refractive powers Q_1 , Q_2 is substituted by a non-astigmatic

eye with power $Q = \frac{Q_1 + Q_2}{2}$. Considering the *virtual wavefront* $\tilde{W}$, we demand that refracting the virtual wavefront $\tilde{W}$ at the non-astigmatic eye results in the same wavefront W'' as refraction of W' at the real, astigmatic eye. Using Equation 6 it is easy to compute the curvatures $\tilde{\kappa}_1$, $\tilde{\kappa}_2$ of $\tilde{W}$:

$$\tilde{\kappa}_i = \frac{\kappa'_i - Q}{1 + b(\kappa'_i - Q)}. \quad (10)$$

Finally, we define $A_{eff} := \tilde{\kappa}_1 - \tilde{\kappa}_2$ and obtain the generalized error functional

$$J(F) := \int_{\Xi} \alpha A_{eff}^2 + \beta (P - P_{des})^2 d\phi d\theta. \quad (11)$$

It is clear by definition that Equation 11 is a true generalization of Equation 9.

In order to accurately evaluate the astigmatism of the lens, we do not simply use a planar incident wavefront W but a wavefront that originates at some distance d from the eye. We vary the distance d between infinity for the far vision area and $d_N = 30\text{cm}$ for the near vision area, interpolating in between depending on the desired refractive power $d = d_N(P_N - P_F) / (P_{des} - P_F)$.

Remark: Traditional approaches^{15, 17} establish error functionals that only consider the geometry of the front surface. These approaches are based on the thin lens formula in Equation 5, fixing an appropriately chosen back face of the lens. This allows us to reduce the optimization to the front face of the lens. The corresponding error functional is

$$J(F) := \int_{\Xi} \alpha (\hat{k}_1 - \hat{k}_2)^2 + \beta \left(\frac{\hat{k}_1 + \hat{k}_2}{2} - H_{des} \right)^2 d\phi d\theta, \quad (12)$$

where H_{des} specifies the distribution of mean curvature on the front face and roughly corresponds to the refractive power of the lens. In some sense, this is a reasonable approximation, since it reduces the optimization task to the front surface. Since the back face is always chosen as a torus, the optimization has to be restricted to the front face anyway. However, Equation 12 is only a rough approximation and does not lead to an optimal solution.

5. Optimization of the error functional

We now describe how the error functional of Equation 11 can be minimized efficiently. In the optical industry, the back face is still always toroidal or spherical. The radii of the torus are derived from the prescribed lens powers P_1, P_2 by means of standard tables and have been chosen so that flat or very highly curved front surfaces are avoided. As a consequence of the fixed back surface, we can restrict the optimization of the PAL to the front surface of the lens.

We start start by introducing a problem specific surface representation for the front of the lens that simplifies and accelerates the numerical evaluation of the functional. Based on this fast evaluation, we also simplify the exact highly non-linear functional and approximate it with a data dependent

quadratic functional, which can be solved much faster using an iterative technique.

5.1. Surface representation

One evaluation of the error functional requires to compute a line-surface intersection for each eye ray. When naively representing the front and back surface (F and B) of the spectacle lens, this has to be done numerically which is very time consuming. We therefore use a surface representation which allows us to compute the intersection points directly.

First we characterize each viewing direction r in polar coordinates, with respect to the center of the eye and with the pole positioned along the viewing direction, that is

$$r = r(\phi, \theta) = (\sin \phi \cos \theta, \sin \theta, \cos \phi \cos \theta). \quad (13)$$

We parameterize the back face B by its distance ρ to the center of rotation of the eye along each viewing direction $r(\phi, \theta)$: $B(\phi, \theta) = \rho(\phi, \theta)r(\phi, \theta)$ (see Figure 15). Analogously, we parameterize the front surface F by the distance σ from the back to the front surface along a refracted ray. Therefore F is parameterized by

$$F(\phi, \theta) = B(\phi, \theta) + \sigma(\phi, \theta)s(\phi, \theta), \quad (14)$$

where s is the normalized direction into which an eye ray is refracted by the back face B .

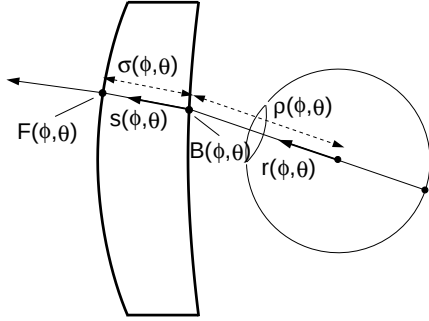


Figure 15: An appropriate parameterization of front and back face of the lens avoids to compute intersections numerically.

Given a fixed back face B , the corresponding distance function ρ needs to be computed once. We evaluate ρ at a finite number (about 100×100) of sample points p_{ij} . Computing each ρ_{ij} does require line-surface intersections, but this can be done in a preprocessing step with negligible computational effort compared to the optimization step.

Since B is fixed in advance, Equation 14 gives a direct correlation between the front surface and the scalar function $(\phi, \theta) \mapsto \sigma(\phi, \theta) \in \mathbb{R}$. Numerical evaluation of the error functional can now be done without computing any line-surface intersections. The optimization itself is reduced to finding an optimal $\sigma^*(\phi, \theta)$ which minimizes Equation 11.

5.2. Minimizing the error functional

Thanks to the chosen surface parameterization, we can formulate our error functional from Equation 11 as $J(F) = J(\sigma) = \int K(L(\sigma))$, where we use the abbreviation $L(\sigma) := (\sigma, \sigma_\phi, \sigma_\theta, \sigma_{\phi\phi}, \sigma_{\phi\theta}, \sigma_{\theta\theta})$. The functional is evaluated numerically by the wavefront tracing techniques described above. We now have to find an optimal $\sigma : \mathbb{R}^5 \rightarrow \mathbb{R}$. Functional J is highly non-linear and minimization by standard techniques, like conjugate gradients, requires high computational effort²².

In contrast, the minimization of quadratic functionals, i.e. functionals where $K : \mathbb{R}^5 \rightarrow \mathbb{R}$ is a quadratic polynomial in $\sigma \dots \sigma_{\theta\theta}$, is relatively easy. In fact, the minimization of quadratic functionals can be reduced to solving a simple linear system of equations.

However, there will be an approximation error due to substituting a quadratic functional for the exact one. In order to avoid this error, we use a quadratic data dependent functional that approximates J with respect to a reference surface σ^i . We then iteratively improve the quality of the reference surface, by using the result of the previous optimization step as our new reference surface¹⁰.

Given a *reference surface* σ^0 as an initial guess, we locally approximate J by its Taylor expansion at σ^0 . The resulting quadratic functional becomes

$$\begin{aligned} J_{\sigma^0}(\sigma) &:= J(\sigma^0) \\ &+ \int \nabla K(\bar{L}(\sigma^0))^t \bar{L}(\sigma - \sigma^0) \\ &+ \frac{1}{2} \int \bar{L}(\sigma - \sigma^0)^t \text{Hess} K(\bar{L}(\sigma^0)) \bar{L}(\sigma - \sigma^0), \end{aligned} \quad (15)$$

where ∇K and $\text{Hess} K$ are again evaluated numerically since no closed form of K is available.

If the initial guess σ^0 is close to the minimum of J , we may hope that the minimum σ^1 of the simplified functional J_{σ^0} is close to the minimum of the exact functional given in Equation 11. By iterating this procedure, we get a sequence $(\sigma^i)_{i \in \mathbb{N}}$, where σ^{i+1} is achieved by minimizing the quadratic functional J_{σ^i} . Starting with a sufficiently accurate initial surface σ^0 , the series $(\sigma^i)_{i \in \mathbb{N}}$ quadratically converges to the minimum of the exact functional.

It can be shown that the above technique converges^{17, 22}. In fact, the algorithm is an implicit application of the well-known Newton-iteration. In practice, we found that $\sigma^0 \equiv \text{const}$ is always a sufficiently good initial guess, leading to a satisfying solution after at most two iteration steps. More iterations did not improve the quality any further. This iterative minimization of the quadratic approximation from Equation 15 took 62.5 seconds on a SGI Onyx with MIPS R10000 processor. We used a bi-cubic tensor product NURBS surface with a total of 1089 control points for representing σ accurately and with enough degree of freedom for the optimization algorithm.

6. Examples

In order to compare the results of the optimization method to currently available lenses we displayed power and effective astigmatism of four PALs in Figure 17. For the purpose of comparing lenses, these plots are more effective than the visual comparison of the images as generated in the first part of this paper.

The lenses on the left hand side are examples for currently available lens designs that are optimized with the usual geometric heuristics. The lenses on the right side are created using the described optimization approach. Their design specifications are similar to those of the respective existing lenses. The optical characteristics of all lenses are -1 dpt for the far vision area and $+1$ dpt for the near vision area.

The existing lens displayed in Figure 17a has tolerable astigmatism, but the refractive power does not increase monotonically from the upper to the lower part; the usable portion of the far vision area is therefore restricted to a small region. In contrast, the lens in Figure 17b has correct refractive power along the whole main viewing line. However, intolerable deviations occur next to the near vision area. Moreover, the technique used to create this lens could not sufficiently reduce the astigmatism.

The lenses of Figure 17c and 17d are created with the method described in this paper. These lenses obviously do not have these problems: the variation of the refractive power from far to near vision area is monotonic and the effective astigmatism is reduced to a satisfactory level.

7. Conclusion and Future Work

We presented two new approaches for visualizing and optimizing progressive addition lenses. Both techniques are based on wavefront tracing for obtaining the properties of an optical system.

A new approach to rendering images by continuously adjusting the eye's accommodation across the field of view results in images that can be used to determine the visual quality of newly designed lenses. The images are the result of simulating the human vision as modified by the optical system. This technique uses wavefront tracing to obtain the required data to focus the eye on the object visible in each direction.

Additionally, we presented a technique for designing optimized progressive lenses by minimizing an error functional that measures the optical aberrations. Evaluation of the error functional is again based on wavefront tracing. In combination with an appropriate surface representation, we obtain a fast method for evaluating the error functional by avoiding line-surface intersection calculations.

The minimization of the error functional relies on approximating the error measure by quadratic functionals, which leads to a quadratically converging series of approximating

solutions, becoming quasi-stationary after only two steps of the iteration. Lenses designed with the new approach are closer to the prescribed properties and exhibit a smoother distribution of refractive power and astigmatism than lenses designed with traditional approaches.

The new method is already being used in industry for designing standard sets of free-form surfaces for the front side of PAL lenses. The back side is later chosen and manufactured according to the other corrections required for a client. Although client-specific PAL designs, where both the front and back of the lens are free-form surfaces, would have better optical properties, they are difficult to manufacture in practice and are thus not yet available. However, the same basic approach to the design could be used for also handling this more difficult situation. It remains to be seen if the additional quality is worth the extra effort. In any case, both the new rendering technique and the optimization method are good candidates for use in other areas of computer graphics and the design of optical systems.

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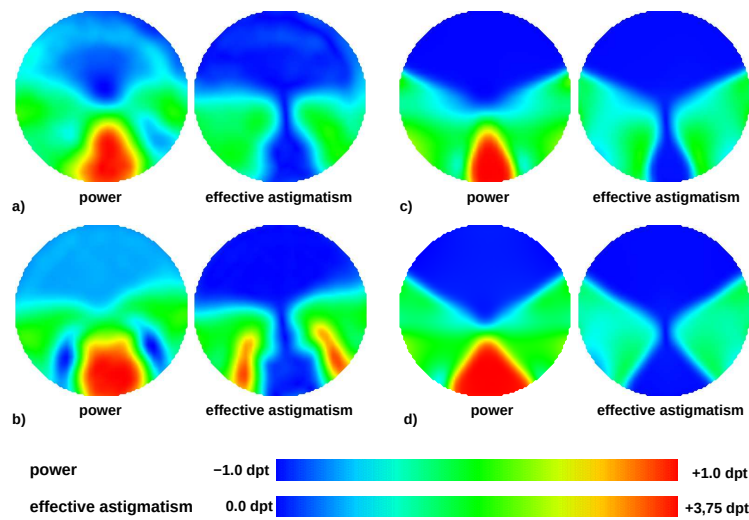
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[Loos, Slusallek, Seidel] **Figure16:** Uncorrected and multiple corrected views: (from left to right) uncorrected vision of a presbyopic eye, correction by uni-focal near vision spectacles, same correction for far vision spectacles, correction by a PAL.



[Loos, Slusallek, Seidel] **Figure17:** The lenses on the left side show numerous defects both in the distribution of refractive power and astigmatism. The optimized lenses on the right side are free from these problems.